Do Fusion Operators Matter, and Does the Answer Survive Preprocessing? A Parameter-Matched Multiverse Comparison of Multimodal CNN Fusion for EEG Seizure Detection

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DRAFT v2, NOT FOR SUBMISSION. Restructured around the three research questions the advisor set on 10 October 2026. This file holds the Introduction (Section 1), the Methods (Section 2, in the advisor's subsections 2.1 to 2.7) and the Results of the five-seed runs (Section 3; the seizure-level metrics of P6 are still being computed) and the Discussion (Section 4). The literature review is provisional: the systematic search has run, but title and abstract screening was done by an AI assistant, and the human screening and the full texts follow after 17 October; claims marked [abstract] rest on a source's abstract only and are to be checked against its full text. `paper/manuscript.md` is the previous draft (Bonn and CHB-MIT, single pipeline) and stays unchanged until this one replaces it; where the Bonn benchmark goes (supplementary material or a separate note) is still to be decided with the advisor. Every number below is traceable to a file in the repository (paths given).

Abstract

Studies of convolutional networks for EEG seizure detection that fuse the raw waveform with its spectrogram often report that one fusion operator outperforms the others, on one dataset, one preprocessing pipeline and one training run. We tested whether such conclusions hold when these choices change. Five fusion operators (early, late, gated, attention and score fusion), with parameter counts matched within 1.3 percent, were compared under nine preprocessing pipelines, including three artefact removal algorithms (Extended Infomax, GEDAI, AMICA), and five seed sets, with leave-one-subject-out evaluation on CHB-MIT (23 subjects) and Siena (14 subjects), and by training on CHB-MIT and testing on Siena. Statistical significance (resampling-corrected paired tests), practical equivalence (margins of 0.05 and 0.02 macro F1 fixed in advance) and ranking stability were assessed separately. The operators differed by a median of 0.021 (CHB-MIT) and 0.053 (Siena) macro F1 within a pipeline, and no difference was significant. The held-out subject explained 74 percent of the variance on CHB-MIT; preprocessing and the operator explained none, and the seed more. Rankings changed between seeds (median Kendall's W 0.22 on CHB-MIT), more stable on Siena, where score fusion ranked first in 82 percent of runs without a significant advantage, and did not transfer between datasets. Early fusion had the lowest AUPRC in all 18 pipeline-dataset combinations. With fixed seeds and package versions, GEDAI cleaned 27 percent of recordings differently on a second machine, enough to change seizure-level sensitivity by up to 0.054 and the order of the operators. Claims that one fusion operator is superior, made from a single dataset, pipeline and run, are not supported.

1. Introduction

Electroencephalography (EEG) records the brain's electrical activity, and during an epileptic seizure that activity changes characteristically: amplitude rises, the signal becomes rhythmic and its power concentrates in particular frequency bands. Automatic seizure detection is therefore a signal classification problem, and convolutional neural networks (CNNs) are the dominant

approach to it. A recurring design represents each EEG segment twice, as the raw waveform for a 1D CNN and as its time-frequency image for a 2D CNN, and combines the two branches with a fusion operator: early fusion of inputs or early features, late fusion of the final feature vectors, a learned gate or attention weighting, or a combination of the two branches' predicted scores. Studies of such models typically report that one operator outperforms the others. Whether such a statement holds once the analytic decisions around the comparison change is the question of this paper.

1.1 Preprocessing changes what deep models achieve

Deep networks are often expected to learn from minimally processed EEG, yet the choice of preprocessing changes their results. In the first systematic study of EEG preprocessing for deep learning, 4,800 models trained on six tasks with four architectures differed significantly between preprocessing pipelines; models trained on raw data ranked last on average, and minimal pipelines without artefact handling tended to serve them best (Del Pup et al. 2025) [abstract]. Artefact removal by independent component analysis, with Infomax or AMICA and three component rejection strategies, did not consistently improve deep decoding on three brain-computer interface datasets (Kang et al. 2024) [abstract], and even the scope of normalisation changed which strategy worked best (Truong and Delorme 2025) [abstract]. Outside deep learning the same sensitivity is documented by multiverse analyses, which run and report every defensible combination of analysis choices instead of one (Steegen et al. 2016; Clayson 2024): across 43 preprocessing pipelines no single pipeline was best on every quality criterion (Huang et al. 2025) [abstract]; fourteen pipelines drawn from the N400 literature changed statistical outcomes, effect sizes and power (Šoškić et al. 2024) [abstract]; and only 13 of 270 analyses of frontal alpha asymmetry in depression were significant (Kołodziej et al. 2021) [abstract]. For event-related potentials, multiverse analyses have been used to choose pipelines by their effect on data quality and reliability (Clayson et al. 2021) [abstract]. A comparison of fusion operators made under one preprocessing pipeline therefore cannot tell whether its ranking belongs to the operators or to the pipeline.

1.2 Fusion operators are compared, but not under controlled conditions

Several recent studies compare fusion operators for EEG seizure detection or prediction directly (Table 1): five strategies within and across patients (An et al. 2026), four operators under leave-one-subject-out evaluation, where they lie within one standard deviation of each other (Einizade et al. 2023), three operators with paired tests over six seeds (Basheer and Mishra 2026), and feature-level against decision-level fusion for seizure prediction (Daşdemir and Örnek 2025). Among the studies we have read in full, none matches the parameter counts of the compared operators, tests the differences between them with a correction for repeated resampling, and tests them for equivalence; a larger fusion model that beats a smaller one may owe its advantage to capacity rather than to the way it fuses. Across more than 120 studies of hybrid EEG interfaces, only two explored more than one fusion strategy and one asked which was best (Lee et al. 2025). Controlled comparisons in other domains point the same way: in ECG classification 1D and 2D representations were compared with early, late and joint fusion (Narotamo et al. 2024), a controlled benchmark on tabular neuroimaging data found the fusion strategy to matter more than the architecture (Camastra et al. 2026), and added multimodal complexity often did not pay off (Rheude et al. 2025).

Table 1: Closest related work (from the systematic search of 3 October 2026, `docs/15_LITERATUR_SONUCLARI.md`; studies in bold are the closest to this one; the full texts of the remaining candidates are still being read).

Work	Contribution	What it leaves open
An et al. (2026), fastSeizureNet	Compares five fusion strategies for EEG seizure detection, within and across patients, with several backbones	Fuses hand-crafted with deep features rather than a waveform with its spectrogram; strategies not parameter matched and not tested against each other (a Wilcoxon test with Holm correction is used only against prior methods)
Einizade et al. (2023), fAttNet	Compares four fusion operators over raw EEG, wavelet packet and hand-engineered views, leave-one-subject-out on TUH	No parameter matching, no statistical test; the operators lie within one standard deviation of each other
Basheer and Mishra (2026)	Compares concatenation, element-wise addition and KAN gating, paired t tests over six seeds	Both streams come from the same raw EEG; uncorrected tests against concatenation only; no parameter matching
Daşdemir and Örnek (2025)	Protocol-controlled comparison of two fusion strategies for EEG	Seizure prediction, not detection; two operators; a 0.20 point accuracy difference without a stated test or parameter matching
Wang et al. (2026), *Neurocomputing*	Cross-attention fusion of time-domain and S-transform views, CHB-MIT and Siena	One ablation against concatenation, patient-specific evaluation, no statistical test
Chatzichristos et al. (2020)	Attention-gated multi-view fusion	One operator, no alternatives
Das et al. (2024); Huang et al. (2024)	1D+2D fusion; multi-head attention fusion	Fixed design, no operator ablation; gains without matched parameter counts
Narotamo et al. (2024)	1D and 2D representations with early, late and joint fusion	ECG; no parameter matching or corrected and equivalence statistics
Camastra et al. (2026), *Brain Sciences*	Controlled fusion benchmark: strategy matters more than architecture	Tabular neuroimaging features; no paired or equivalence testing
Ali et al. (2024)	CHB-MIT results depend heavily on evaluation choices	Not about fusion operators
Dan et al. (2024), SzCORE	Standard datasets, event-based scoring and metrics for seizure detection	An evaluation framework
Kontras et al. (2026), NeuroAtlas	Bonn is saturated (eleven models at AUROC >= 0.99) and has no subject identifiers	Not a fusion comparison

1.3 One seed or one pipeline is a weak basis for a ranking

A comparison built on one training run can be decided by factors unrelated to the methods compared. Identical training runs of deep networks produce different models: even after weak models were excluded, accuracy differed by up to 10.8 percent between identical runs, implementation-level factors alone caused differences of up to 2.9 percent, and only about one fifth of the surveyed papers used repeated runs to quantify this (Pham et al. 2020) [abstract]. Every source of nondeterminism in network optimisation, down to a one-bit change in the initial parameters, produces similar run-to-run variability, which makes small improvements hard to separate from noise (Summers and Dinneen 2021) [abstract], and seeds that perform far better or worse than average are easy to find (Picard 2021) [abstract]. The computing environment adds its own variance: CPU multithreading makes training nondeterministic (Xiao et al. 2021) [abstract], floating-point non-associativity in parallel computation causes run-to-run differences

(Shanmugavelu et al. 2024) [abstract], and reproducible training requires control of both software randomness and hardware nondeterminism (Chen et al. 2022) [abstract]. In speech recognition, training variance was large enough to call for a change in how results are reported (van den Berg et al. 2017) [abstract], and seed-induced variability can be tested for directly (Banerjee et al. 2025) [abstract]. The statistics add a further trap: under repeated cross-validation the training sets overlap, so standard paired tests are overconfident unless their variance is corrected (Nadeau and Bengio 2003; Bouckaert and Frank 2004), and whether a comparison comes out significant can depend on the number of folds and repeats (Jafrasteh et al. 2025).

1.4 Results on one dataset rarely carry over to another

High accuracies reported for automatic seizure detection often do not hold on new patients or recording sites. When 28 algorithms from the SzCORE challenge were evaluated on 4,360 hours of held-out continuous EEG from 65 subjects, the best F1 score was 32 percent, far below self-reported results, and the algorithms with the highest aggregate score were not the most consistent across subjects (Dan et al. 2025, preprint) [abstract]; the same group proposed standard datasets, cross-validation strategies and event-based metrics to make such comparisons possible (Dan et al. 2024). Real-time detectors had been tested in distinct settings until they were compared under one (Lee et al. 2022, preprint) [abstract]; a prospective test on about 14,590 hours of adult EEG from another continent reported 76.68 percent with about 56 false alarms per 24 hours (Yang et al. 2021, preprint) [abstract]; and a systematic review names generalisability across training data and acquisition hardware among the main obstacles to clinical translation (Moutonnet et al. 2024, preprint) [abstract]. Within one public corpus, CHB-MIT results depend heavily on evaluation choices (Ali et al. 2024), and the Bonn corpus, long a standard benchmark, is saturated and carries no subject identifiers, so it cannot test generalisation across patients (Kontras et al. 2026).

1.5 This study

Our contribution is not another fusion operator, nor only a comparison of existing ones: it is a test of how far conclusions drawn from such a comparison hold when the analytic decisions around it change. We hold the backbone, the training protocol and the parameter budget fixed (within 1.3 percent), vary only the fusion operator, and repeat the whole comparison under nine preprocessing pipelines, including three artefact removal algorithms, and five seed sets, on two independent corpora and from one to the other. We ask three questions.

- **RQ1, performance robustness.** Do fusion operators maintain their predictive performance across different EEG preprocessing pipelines?
- **RQ2, ranking robustness.** Are the rankings of fusion operators stable across preprocessing choices, artefact removal algorithms and stochastic training?
- **RQ3, cross-dataset generalisation.** Do fusion performance patterns and rankings generalise across independent EEG seizure datasets?

Throughout, three properties are kept apart, because each can hold without the others: **statistical significance** (is there evidence of a difference between two operators?), **practical equivalence** (does their difference lie within a margin fixed in advance?) and **ranking stability** (is their order preserved when preprocessing or seeds change?). In particular, the absence of a significant difference is not taken as evidence of equivalence.

The study contributes:

1. a parameter-matched comparison of five fusion operators, with single-branch and classical controls, crossed with nine preprocessing pipelines and five seed sets on CHB-MIT and Siena, and a transfer experiment from one to the other;
2. an analysis that separates the effects of preprocessing, seed and their interaction with the operator, and reports each operator's full rank distribution rather than a single ranking;
3. a test of the reproducibility of three artefact removal algorithms across machines and execution contexts, and of how much their irreproducibility changes the comparison itself;
4. corrections to the source data and its documentation: a labelling error in CHB-MIT's annotations, two of its cases that are one person and must be held out together, a mains notch that has to follow each dataset's line frequency, and six problems in Siena's seizure lists, each resolved and documented;
5. the code, the per-fold results of every run, and every analysis, public and regenerated by single commands.

2. Methods

Figure 1 (`paper/figures_manuscript/figure1_design.png`) summarises the design: the datasets, the nine preprocessing pipelines, the two representations, the fusion operators and controls, and the evaluation; its panel b shows where each step is applied and how leakage between training and test data is prevented.

2.1 Datasets and EEG harmonisation

CHB-MIT (Shoeb 2009; Goldberger et al. 2000) holds 686 scalp recordings, 982.9 h, from 24 paediatric cases at 256 Hz in a bipolar montage. Two cases, chb01 and chb21, are documented as the same person and are treated as one subject, which gives 23 subjects. We use the standard 18-channel bipolar (double banana) montage and keep the 683 recordings that contain all 18 channels, with 185 of the 198 annotated seizures in the windowed corpus; the three dropped recordings belong to part of subject chb12's data recorded in a different montage. Seizure annotations were parsed by order rather than by their printed number, which corrects a labelling error in `chb09_08.edf` (`src/chbmit.py: parse_summary`).

Siena Scalp EEG (Detti et al. 2020) holds 41 recordings, 141.0 h by the EDF headers, from 14 adult patients, with 47 annotated seizures, recorded at 512 Hz in a referential 10-20 montage (older electrode names T3 to T6 are mapped to T7, T8, P7, P8). Six problems in the published seizure lists (a seizure ending after its recording, two conflicting onset or end times, two start times that disagree with the EDF headers, an onset given only clinically) were resolved before any analysis, each as either a correction verified against the recording or a stated decision rule; Supplementary Table S4 lists every decision, how it was checked, its remaining uncertainty, and how much the alternative would change the windows (`docs/SIENA.md`).

Harmonisation. Siena is converted to the CHB-MIT form before anything else: the 18 bipolar channels are formed as electrode differences of the referential signals, and the continuous recording is resampled from 512 to 256 Hz with a polyphase filter (`scipy.signal.resample_poly`, factor 1/2; `src/siena.py: read_bipolar`). Both corpora then pass through identical code. Seizure times are converted to seconds from each recording's first sample using the EDF header start time (modulo 24 h for recordings past midnight).

Windows and labels. Recordings are cut into non-overlapping 10 s windows. A window is ictal if it overlaps an annotated seizure at all. Non-ictal windows are subsampled per subject to four per ictal window of that subject, with a fixed seed, and every subject's true recording duration is kept for the event metrics (Section 2.5). This gives 6,380 windows (1,276 ictal) for CHB-MIT

and 1,640 windows (328 ictal) for Siena. The same windows are used by every pipeline, which is checked whenever a pipeline's corpus is built, so all comparisons between pipelines are paired window by window.

Differences between the datasets. The two corpora differ in age (mostly children against adults), seizure types, recording length per patient (Siena's recordings are cut around seizures for a seizure prediction study), centre, equipment and mains frequency (60 Hz in Boston, 50 Hz in Italy). Results are compared between them as an independent replication and a transfer test (RQ3), not interpreted as an age or population effect.

2.2 Preprocessing multiverse

Following the multiverse approach (Steegen et al. 2016), nine defensible preprocessing pipelines are crossed with every model, operator and seed instead of choosing one. Each pipeline changes one hypothesis-driven aspect; windows, labels, spectrogram settings, models, training and evaluation are fixed (`src/preprocess.py`).

Pipeline	Band-pass	Notch	Artefact handling	Normalisation	Question
P0	none	none	none	z-score	original benchmark
P1	0.5 to 40 Hz	none	none	z-score	standard EEG band
P2	0.5 to 70 Hz	mains (CHB-MIT 60 Hz, Siena 50 Hz)	none	z-score	information above 40 Hz
P3	1 to 40 Hz	none	none	z-score	low frequency content and drift
P4	0.5 to 40 Hz	none	technical artefact rejection (training only)	z-score	artefact rejection
P5	0.5 to 40 Hz	none	none	median and IQR	normalisation
P6a	0.5 to 40 Hz	none	Extended Infomax ICA	z-score	artefact removal
P6b	0.5 to 40 Hz	none	GEDAI	z-score	artefact removal
P6c	0.5 to 40 Hz	none	AMICA	z-score	artefact removal

Filters. Zero-phase Butterworth filters (order 4, forward and backward) are applied to each continuous recording before it is windowed, so that no filter transient falls at a window edge. P2's notch removes each dataset's own mains frequency and its first harmonic; with a fixed 60 Hz notch Siena's P2 windows kept a 50 Hz line 318 times stronger than the 40 to 48 Hz background, so P2 was defined per dataset and its Siena runs were repeated.

Normalisation. Each channel of each window is standardised on its own: z-score (P0 to P4, P6) or median and interquartile range (P5). No statistic is computed across windows, recordings or persons, so normalisation cannot carry information from test to training data.

P4, technical artefact rejection. Amplitude was measured before a criterion was chosen: every amplitude threshold removes seizures selectively (500 μV rejects 72 percent of CHB-MIT's ictal and 48 percent of its non-ictal windows), because ictal windows have larger amplitudes. P4 therefore rejects only technical artefacts: a flat channel (standard deviation below 1 μV in the window) or a constant run of at least 0.5 s on any channel (a signal dropout), detected on the unfiltered signal. In CHB-MIT this rejects 42 of 1,276 ictal (3.3 percent) and 39 of 5,104 non-ictal

windows (0.8 percent); the difference reflects which recordings have dropouts, not physiology (Supplementary Table S3). In Siena it rejects one window. Rejected windows are removed from training and validation only; the test windows are those of every other pipeline.

P6, three artefact removal algorithms. P6 applies Extended Infomax ICA (Lee et al. 1999; MNE-Python, Gramfort et al. 2013), GEDAI (Ros et al. 2025) or AMICA (Palmer et al. 2008; `jamica`) to the same 0.5 to 40 Hz signal, recording by recording. Infomax and AMICA share every setting and one component rejection rule: 16 components (the 18-channel bipolar montage has rank 16, because it contains two closed electrode loops per hemisphere), fitted on every 8th sample, with components removed when their correlation with any of the four frontopolar channels, used as EOG proxies since neither dataset has an EOG channel, has $|z| > 3$ (MNE `find_bads_eog`). Muscle components are not removed, because the gamma band carries the strongest ictal signal in CHB-MIT. GEDAI is not an ICA: it removes what departs from a leadfield-based reference covariance, by its own criterion, so the shared rule cannot apply. Its reference covariance C, defined for a referential layout, is mapped onto the bipolar montage as $D\,C\,D^{\top}$ (D the electrode difference matrix), and the channel mean is removed before and restored after cleaning. GEDAI's threshold preset was fixed before any classification run, on signal preservation alone: on six CHB-MIT subjects the default removed more than half of the power of 39 percent of recordings, the conservative preset of 17 percent, and neither removed more from ictal than from non-ictal windows (Supplementary Table S2); P6b uses the conservative preset. The comparison among P6a, P6b and P6c is therefore a comparison of whole cleaning procedures: for Infomax and AMICA it isolates the decomposition algorithm, while GEDAI differs also in how it decides what to remove. Signal retention and compute are in Supplementary Table S1, and the power removed by band and class in Supplementary Table S5.

No leakage through P6. Each recording is decomposed and cleaned on its own, from its own unlabelled signal; seizure annotations are never used, and no decomposition, threshold or statistic is shared between recordings or persons. A test person's recordings are therefore cleaned exactly as they would be without any training data.

2.3 EEG representations and fusion strategies

Representations. Each 10 s window enters the network twice: as the 18-channel waveform (2,560 samples per channel) and as its log-magnitude short-time Fourier transform, $\log(1 + |\mathrm{STFT}|)$, computed per channel with a 1 s Hann window, 0.5 s hop and 256-point FFT and cropped at 64 Hz (65 frequency bins by 19 frames; `src/chbmit_prep.py`). The 64 Hz ceiling is kept in every pipeline, so input shapes and parameter counts never change.

Architecture. A 1D convolutional encoder (channel widths 16, 32, 64) processes the waveform and a 2D convolutional encoder of matching depth the spectrogram; each yields a 128-dimensional pooled feature vector. Five interchangeable fusion operators combine them:

- **Early:** the waveform passes a small strided 1D encoder, is broadcast along the spectrogram's time axis and concatenated to it as an extra channel before the 2D backbone (architecturally an intermediate fusion, since the two inputs share no tensor shape).
- **Late:** the two feature vectors are concatenated and classified by a small MLP.
- **Gated:** a learned scalar gate per branch reweights the concatenated features.
- **Attention:** a learned attention module weights the branches from both feature vectors jointly.
- **Score:** the two branches are trained as independent classifiers and their probabilities blended with a weight chosen on the validation split only.

Parameter matching. Every jointly trained model (the four joint operators and two widened single-branch controls) is sized to a common budget by solving the fusion head's hidden width

numerically (`src/models.py`): with 18 input channels the models have 102,236 to 103,712 parameters, the four joint operators within 0.36 percent of each other. Score fusion, an ensemble of two independently sized classifiers, keeps its natural size.

Controls. The two single-branch networks at their default width (raw1d, spec2d) and widened to the fusion budget (raw1d_wide, spec2d_wide), which asks whether fusion beats a unimodal model of the same size; and two classical baselines without representation learning, a logistic regression on log variance (logvar) and on seven hand-crafted features (log variance, line length and relative power in five bands; shallow).

2.4 Experimental protocol

Leave one subject out. Each subject is held out once as the test set (23 folds on CHB-MIT, chb01 and chb21 together; 14 on Siena). From the remaining subjects, whole subjects are set aside as an inner validation set (4 of 22 on CHB-MIT, 3 of 13 on Siena), chosen deterministically to cover the range of seizure burdens; it serves early stopping and the score fusion weight only. No subject's windows and no seizure's windows ever appear on both sides of a split, which a leakage checker confirms for every split and which was itself validated on a deliberately leaky split (`src/chbmit_corpus.py`).

Training. AdamW (Loshchilov and Hutter 2019), learning rate 1e-3, weight decay 1e-4, batch size 32, class-weighted cross-entropy; early stopping on validation macro F1 with patience 12 after a minimum of 20 epochs, at most 60 epochs. The minimum guards against a model stopping in a trivial single-class solution, found in an early audit.

Seeds. Every pipeline is run with five seed sets. A run's seed for each fold and model is derived from the seed set, fold and model (`src/config.py: fold_seed`), so it does not depend on execution order. The ranking of the operators was unstable across three seed sets (Kendall's W below 0.5 in seven of nine CHB-MIT pipelines), the condition fixed beforehand for extending to five.

Cross-dataset experiment (RQ3). Each model is trained once on all CHB-MIT subjects (with the same inner validation rule) and tested on every Siena window and recording, for every pipeline and seed set (`src/cross_dataset.py`). The pipeline is applied to each dataset as defined, including each dataset's own mains notch in P2. Results are reported per Siena patient.

2.5 Evaluation metrics

Window level (primary). Macro F1 and the area under the precision-recall curve (AUPRC) are primary; balanced accuracy (the mean of sensitivity and specificity) is reported alongside, with AUROC and the Brier score as secondary. A window is predicted ictal when its ictal probability exceeds 0.5. Metrics are computed per fold on the held-out subject's windows and averaged over folds within a run.

Seizure level (secondary, clinical). Every model saved for a fold is applied to every 10 s window of every recording of the held-out subject, not only the subsampled test windows, after the pipeline's own preprocessing; a window is positive when its probability exceeds 0.5, with no further post-processing. Detections are scored against the annotated seizures with SzCORE's event scoring (Dan et al. 2024; `timescoring` 0.0.7, defaults: 30 s tolerance before onset and 60 s after offset, any overlap counts, events closer than 90 s merged, events longer than 5 min split). Seizure-level sensitivity is the share of seizures detected; the false alarm rate is false-positive events per hour on the true recording time. The scored recordings hold 187 seizures in 979.9 h (CHB-MIT) and 47 seizures in 141.0 h (Siena); scoring the window labels themselves as predictions detects all of them without a false alarm, which checks the reference, the windows

and the scoring together. On Siena only sensitivity is reported: its recordings are cut around seizures, so a false alarm rate would not describe continuous monitoring. The criterion, the merging of consecutive detections and the counting of false alarms were fixed, and committed, before any multiverse run (`docs/EVENTS.md`), and are the same for every operator and pipeline.

2.6 Statistical analysis

Pairwise comparisons. Leave-one-subject-out folds share training data, so the standard paired test overstates significance; in a simulation of two equivalent classifiers the naive paired Wilcoxon test rejected in 37.6 percent of cases at a nominal 5 percent (`src/sim_cv_correlation.py`). Pairs of operators are therefore compared with the resampling-corrected paired t test (Nadeau and Bengio 2003; test to training ratio 1/22 on CHB-MIT, 1/13 on Siena), with Holm correction over the ten operator pairs within each pipeline.

Practical equivalence. For each pair the two one-sided test with the same corrected variance gives the smallest margin within which the pair is equivalent at alpha = 0.05; a pair is equivalent if that margin is at most 0.05 macro F1 (main analysis) and, as a stricter sensitivity analysis, 0.02. Both margins were fixed and committed before any multiverse result existed. A pair is robustly equivalent if it is equivalent in every pipeline.

Ranking stability. In every run (pipeline and seed set) the five operators are ranked by mean macro F1. We report each operator's full rank distribution (counts per rank, median, best and worst rank, share of runs won and lost), Kendall's W across seed sets within each pipeline, and Kendall's tau between the seed-averaged rankings of every pair of pipelines. Whether preprocessing changes the ranking more than the seed does is tested by comparing the mean tau between seed sets of the same pipeline with the mean tau between pipelines, against a null distribution from 10,000 permutations of the pipeline labels within each seed set; the same test is run among P6a, P6b and P6c alone.

Fusion x Preprocessing. A repeated-measures ANOVA on macro F1 (seed means) treats the held-out subjects as the repeated units and pipeline and operator as within-subject factors; each effect is tested against its interaction with subject, with Greenhouse-Geisser correction. Effect sizes are reported as partial eta squared.

Variance components. For macro F1, AUPRC and balanced accuracy, the fully crossed subject x pipeline x operator x seed design is decomposed as y = mu + subject + pipeline + operator + seed + every interaction, all treated as random effects (generalizability theory) and estimated by the method of moments from the expected mean squares. With one observation per cell the four-way interaction is the residual. Every component is reported unrounded, as an estimated variance, as a standard deviation in the metric's own units (a practical size) and as a share of the total; negative estimates, which arise when an effect's mean square is smaller than its error term, are reported as such and set to zero in the shares.

Prediction stability. The agreement (Cohen's kappa) of window-level predictions between pipelines at the same seed set is compared with the agreement between seed sets in the same pipeline.

Statistical significance is reported from the corrected tests and the ANOVA, practical size from the equivalence margins and the variance components in metric units, and ranking stability from the rank analyses; a result in one of these three is never used as evidence for another. All analyses are implemented in `src/multiverse.py` and were validated on synthetic runs with known effects (39 checks).

2.7 Reproducibility and computational cost

Software and seeds. Training used Python 3.12, numpy 1.26.4 and PyTorch 2.8.0 (CUDA 12.8 build); artefact removal numpy 2.5.3, MNE 1.13.2, GEDAI 0.60 and jamica 0.3.0, in a separate environment because they need a newer numpy (`requirements.txt`, `requirements-ica.txt`). Seeds are fixed per run, fold and model, and per recording for artefact removal (from the file name).

Determinism and device. All runs of one analysis use one machine, one device and 16 CPU threads. On the CPU a run was reproduced bit for bit, also with two runs sharing the machine. The multiverse ran on one GPU (NVIDIA RTX 5070) with PyTorch's deterministic algorithms; adaptive average pooling has no deterministic CUDA backward, yet two GPU runs of the same fold were identical. Before switching to the GPU, its agreement with the CPU was tested at identical seeds against the difference caused by the seed alone: over five folds the mean absolute macro F1 difference between CPU and GPU was 0.074, and between two CPU seed sets 0.066, so the GPU behaves as one more seed rather than as a systematic shift (`docs/PIPELINE.md`). CPU and GPU runs are never mixed in one analysis.

Reproducibility of artefact removal. Fixing seeds and package versions did not make P6 reproducible across machines and execution contexts (`docs/ICA_REPRO.md`). AMICA reproduced up to rounding. Extended Infomax removed different components in recordings whose component lay near the $|z| = 3$ rule: 2.5 percent of recordings between two machines and 1.2 percent between two runs on the same machine with different process parallelism. GEDAI was deterministic on each machine but gave a different result on the second machine in 27 percent of recordings, often by a large amount. A third machine of another architecture (ARM) agreed with the other two on recordings where they agreed, and often with neither where they differed. Multiplying the input by (1 + 1e-7 N(0, 1)), a rounding-level perturbation, changed GEDAI's result on exactly the recordings that differed between machines, also on a full-rank montage, and Infomax's component decision on recordings where it had differed; AMICA was unchanged. The differences therefore come from the methods' sensitivity to rounding near a decision threshold on particular recordings, not from one machine or library. Window-level results are unaffected, because every P6 training and test window comes from one set of signals, cleaned once. The event metrics, which need every recording cleaned again, were computed on the machine that produced those signals; they then reproduce the training-time cleaning exactly for GEDAI, within 1.4 percent of the amplitude for AMICA, and differ for Infomax in 8 of 670 CHB-MIT recordings (`events/meta.json`, `window_check`).

Computational cost. Artefact removal of the 670 CHB-MIT recordings with selected windows took 4.7 (GEDAI), 8.5 (Infomax) and 73 (AMICA) CPU hours on a laptop CPU (Supplementary Table S1). One CHB-MIT run (23 folds, nine models) took about 1.5 h on the GPU, one Siena run about a quarter of an hour, and one cross-dataset run about 5 min; one fold took 3.7 min on the GPU against 12 min on the CPU.

Availability. Code, the per-fold results of every run, and all analyses are in the public repository; Section 9 lists the commands that regenerate each table and figure. Following our own recommendation (Section 4.5), the repository also holds, per recording and artefact removal method, the outcome of the cleaning the models were trained and tested on (components removed, GEDAI's threshold, power kept; `results_v2/qc/p6_recording_logs.csv`) and of the recleaning used for the event metrics (`p6_recording_logs_events.csv`), so that a reader can check a reproduction recording by recording.

3. Results

All numbers below come from the 45 runs per dataset (nine pipelines by five seed sets; 23 folds on CHB-MIT, 14 on Siena; `results_v2/multiverse/chbmit_cuda`, `results_v2/multiverse/siena_cuda`) and the 45 cross-dataset runs (`results_v2/cross`). Means over runs are given with the standard deviation over runs.

3.1 Performance across preprocessing pipelines (RQ1)

Table 2: Window-level performance of the fusion operators, mean and SD over the 45 runs of each dataset (fold means within a run). Per pipeline: Figure 2 and Supplementary Table S6.

Operator	CHB-MIT macro F1	CHB-MIT AUPRC	CHB-MIT bal. acc.	Siena macro F1	Siena AUPRC	Siena bal. acc.
early	0.677 ±0.013	0.613 ±0.024	0.682 ±0.011	0.617 ±0.021	0.496 ±0.035	0.642 ±0.022
late	0.681 ±0.015	0.656 ±0.019	0.679 ±0.012	0.640 ±0.025	0.599 ±0.040	0.679 ±0.018
gated	0.686 ±0.016	0.666 ±0.018	0.683 ±0.013	0.631 ±0.031	0.591 ±0.034	0.674 ±0.022
attention	0.683 ±0.014	0.655 ±0.020	0.682 ±0.013	0.625 ±0.023	0.580 ±0.034	0.667 ±0.020
score	0.673 ±0.017	0.655 ±0.017	0.678 ±0.015	0.670 ±0.024	0.631 ±0.033	0.689 ±0.022

The operators performed within a narrow band. On CHB-MIT the best and worst operator of a pipeline differed by 0.013 to 0.035 macro F1 (median 0.021), and each operator's mean varied across the nine pipelines with an SD of 0.007 to 0.008. On Siena the spread was larger, 0.042 to 0.070 within a pipeline (median 0.053) and an SD of 0.011 to 0.024 across pipelines. Preprocessing had no detectable main effect: in the repeated-measures ANOVA the pipeline effect was $F = 0.29$ on CHB-MIT (Greenhouse-Geisser $p = 0.83$, partial eta squared 0.013) and $F = 1.09$ on Siena ($p = 0.36$, 0.077), and its variance component was estimated below zero on CHB-MIT (-2.7e-5) and at 0.06 percent of the variance on Siena (SD 0.003 macro F1).

One pattern held in every pipeline of both datasets: early fusion had the lowest AUPRC, in 9 of 9 pipelines on CHB-MIT (0.564 to 0.632 against at least 0.631 for every other operator) and in 9 of 9 on Siena (0.453 to 0.540 against at least 0.522), although its macro F1 at the 0.5 threshold was close to the others on CHB-MIT. Its ranking of windows by ictal probability is therefore consistently the weakest, while its thresholded decisions are not.

3.2 Statistical significance and practical equivalence

No pair of operators differed significantly in any pipeline of either dataset: 0 of 90 pipeline-pair comparisons per dataset after Holm correction (smallest corrected p 0.19 on CHB-MIT, 0.095 on Siena). The absence of a difference is not equivalence. On CHB-MIT, 73 of the 90 comparisons were equivalent within 0.05 macro F1 and two pairs, late-attention and gated-attention, were equivalent in every pipeline; within the strict 0.02 margin only 1 of 90 was. On Siena, with fourteen subjects, 21 of 90 were equivalent within 0.05, none in every pipeline, and none within 0.02. The smallest margin at which equivalence could be shown had a median of 0.040 on CHB-MIT (0.020 to 0.066) and 0.081 on Siena (0.021 to 0.148): the data can exclude large differences between operators on CHB-MIT, but not differences of 0.02, and on Siena not even many of 0.05.

3.3 Ranking stability (RQ2)

Figure 3 shows the rank of every operator in every run. On CHB-MIT no operator held its place: gated fusion ranked first in 38 percent of runs, attention in 24, late in 20, early in 11 and

score in 7, and every operator was last in at least 13 percent of runs. Changing only the seed reordered the operators almost completely: Kendall's W across the five seed sets of one pipeline had a median of 0.22 (0.14 to 0.62 over pipelines), and two rankings from different seeds of the same pipeline agreed with a mean Kendall tau of 0.10. Rankings from different pipelines agreed with a mean tau of 0.03; the difference, 0.07, did not exceed what permuting the pipeline labels produces ($p = 0.073$), so preprocessing did not change the ranking more than the seed did. The same held among the three artefact removal pipelines alone (difference -0.07, $p = 0.88$).

On Siena the ranking was more stable. Score fusion ranked first in 82 percent of runs and never last (mean rank 1.29); Kendall's W had a median of 0.58 (0.40 to 0.78), seed rankings agreed with a mean tau of 0.37 and pipeline rankings with 0.34 ($p = 0.074$). Among the three artefact removal pipelines, however, the ranking changed more between algorithms than between seeds (difference 0.18, $p = 0.029$). Score fusion's consistent first place did not amount to a significant difference (Section 3.2): its advantage over the next operator, 0.030 macro F1 on average, lies within the range that fourteen subjects can neither confirm nor exclude, and the operator main effect was $F = 2.87$ (uncorrected $p = 0.032$, Greenhouse-Geisser $p = 0.092$, partial eta squared 0.18).

Fusion x Preprocessing. On CHB-MIT the interaction was $F = 1.73$ with an uncorrected $p = 0.008$ but strong non-sphericity (Greenhouse-Geisser epsilon 0.34), which gives $p = 0.069$; partial eta squared was 0.073 and the interaction's variance component 0.09 percent (SD 0.004 macro F1). With three seed sets the same test had given $p = 0.034$; the evidence for an interaction did not survive the extension to five. On Siena there was no interaction ($F = 1.05$, $p = 0.39$). The window-level predictions themselves changed about as much between pipelines as between seeds: Cohen's kappa 0.56 between pipelines and 0.58 between seeds on CHB-MIT, 0.44 and 0.50 on Siena.

3.4 Sources of variance

Figure 4 decomposes the variance of each metric. On CHB-MIT the held-out subject accounted for 74.2 percent of the variance of macro F1 (SD 0.126), the interaction of subject with pipeline and operator for 7.2 percent and the residual for 17.3 percent; all seed terms together accounted for 1.1 percent (SD 0.015), the Fusion x Preprocessing interaction for 0.09 percent, and the estimates for pipeline and operator were negative (-2.7e-5 and -2.4e-5), that is, below their error terms. AUPRC (subject 76.8 percent, operator 0.8, seed 1.5) and balanced accuracy (72.9, 0, 1.4) showed the same structure. On Siena the subject accounted for 32.5 percent and its interactions with pipeline and operator for 26.9 percent; the operator held 1.5 percent of the variance of macro F1 (SD 0.016) and 5.5 percent of AUPRC, the seed 4.2 percent and preprocessing 0.06 percent. In both datasets the seed accounted for more variance than preprocessing, and on CHB-MIT more than the operator.

3.5 Artefact removal

Figure 5 summarises the three artefact removal pipelines. Infomax and AMICA removed a median of one component per recording and kept a median of 78 percent of the signal's power; GEDAI kept 99.9 percent in most recordings but removed more than half of the power in 21 percent of them. AMICA cost 73 CPU hours on the 670 CHB-MIT recordings, Infomax 8.5 and GEDAI 4.7. Against P1, the same signal without artefact removal, the change in macro F1 was small and depended on the operator: on CHB-MIT late fusion gained 0.014 to 0.022 under all three algorithms and early fusion lost 0.012 to 0.022; on Siena GEDAI raised late and gated fusion by 0.045 and 0.061. AUPRC fell under every algorithm for most operators, by up to 0.060 (early fusion under GEDAI, CHB-MIT).

What the methods remove (Supplementary Table S5): Infomax and AMICA removed mostly delta power (a median of 15 to 18 percent of it on CHB-MIT) and, within a recording, slightly less from ictal than from non-ictal windows (a median difference of 0.010 and 0.017 of the power kept, Wilcoxon $p < 0.001$); GEDAI's conservative preset removed almost nothing on CHB-MIT and showed no difference between the classes within recordings. Over the whole corpus, however, GEDAI's default preset kept a median of 63 percent of the power of CHB-MIT's ictal windows against 93 percent of the non-ictal ones, removing mainly delta and theta power, because the recordings it cleaned heavily held many of the seizures; within recordings it showed no difference. On Siena the conservative preset behaved the same way over the corpus (68 against 99.7 percent) without a difference within recordings ($p = 0.68$). A cleaning step that treats the classes alike within each recording can still remove ictal signal selectively from the corpus as a whole.

Artefact removal was not reproducible across machines (Section 2.7, `docs/ICA_REPRO.md`): GEDAI cleaned 27 percent of recordings differently on the second machine, Infomax changed its component decision in 2.5 percent, AMICA in none. On the recordings that differed, a rounding-level perturbation of the input on one machine changed the result in the same way, and a third machine often gave a third result. Scoring the same models on recordings cleaned on the two machines changed seizure-level sensitivity by up to 0.054 and the operators' order by sensitivity (Kendall tau 0.60 to 1.00) for GEDAI, by at most 0.005 for Infomax, and not at all for AMICA.

3.6 Seizure-level detection

(P0 to P5, five seed sets each; P6 follows when its remaining event metrics are computed.) On CHB-MIT the operators detected 68 to 80 percent of seizures on average, with 5.3 to 7.8 false alarms per hour at the 0.5 threshold without post-processing (`results_v2/results/events_chbmit.csv`). Early fusion detected the most seizures (mean 0.80, the highest in 83 percent of runs) but also raised the most false alarms (7.8 per hour against 5.3 to 5.7 for the others): at the common threshold it labels more windows ictal, consistent with its weaker probability ranking (Section 3.1). On Siena the operators detected 83 to 86 percent of seizures, with no consistent order.

3.7 Cross-dataset generalisation (RQ3)

Models trained on all of CHB-MIT and tested on Siena reached a macro F1 of 0.641 (attention) to 0.677 (late) on average over pipelines and seeds (Figure 6a), higher than the same operators trained on Siena itself for early, late, gated and attention fusion, and lower for score fusion (0.652 against 0.670). The ranking did not transfer. Score fusion, first in 82 percent of Siena's own runs, was first in 11 percent of the transfer runs, where late fusion led (38 percent) and attention was most often last (40 percent). Between the seed-averaged rankings of the same pipeline, Kendall tau was -0.11 on average between CHB-MIT and Siena, 0.07 between CHB-MIT and the transfer, and -0.11 between Siena and the transfer (Figure 6b): neither dataset's ranking predicted the other's, nor the ranking after transfer.

4. Discussion

We asked whether conclusions about fusion operators for EEG seizure detection hold when the preprocessing, the artefact removal algorithm, the training seed and the dataset change. On both datasets the answer is largely no: the operators performed within a few hundredths of macro F1 of each other, no difference between them was significant, and their order changed with the

seed, with the artefact removal algorithm on Siena, and between datasets. We take the advisor's questions in turn.

4.1 How large are the differences between fusion operators?

Small. Within a pipeline the best and worst operator differed by a median of 0.021 macro F1 on CHB-MIT and 0.053 on Siena, and no pairwise difference survived the corrected test. The operator's share of the variance was estimated below zero on CHB-MIT and at 1.5 percent on Siena. One operator did stand apart in one respect: early fusion ranked windows by ictal probability worst in every pipeline of both datasets (lowest AUPRC in 18 of 18 cases), while its thresholded macro F1 was unremarkable and its seizure-level sensitivity, at the price of more false alarms, the highest. Early fusion is the only operator that combines the two representations before either branch has learned its own features, and its weaker ranking of windows is the one property of an operator that the multiverse preserved. Whether two operators are *equivalent* is a separate question, and the answer depends on the dataset: on CHB-MIT most pairs were equivalent within 0.05 macro F1 in most pipelines and two in all, but within 0.02 almost none, and on Siena few were equivalent even within 0.05. The data therefore rule out large differences between operators on CHB-MIT, but not differences of the size that published comparisons typically report as improvements.

4.2 Are the differences preserved when preprocessing changes?

Preprocessing had no detectable main effect on performance and did not reorder the operators more than a change of seed did. The interaction between operator and preprocessing, significant with three seed sets on CHB-MIT, was no longer significant after correction with five ($p = 0.069$), and its variance component was 0.09 percent. The one place where preprocessing mattered more than the seed was the choice among the three artefact removal algorithms on Siena ($p = 0.029$), where GEDAI raised late and gated fusion by about 0.05 macro F1. This agrees with the wider literature, where preprocessing changes results but no pipeline is best throughout (Section 1.1), and it adds that, for a comparison of operators, the pipeline mostly adds noise rather than a systematic advantage for one operator.

4.3 How much does the random seed change the rankings?

More than preprocessing does. On CHB-MIT the rankings from five seeds of the same pipeline agreed barely above chance (median Kendall's W 0.22), all seed terms together explained more of the variance than preprocessing and the operator, and the window-level predictions changed as much between seeds as between pipelines. A ranking obtained from one training run is therefore close to a random draw on CHB-MIT. The consequence for the literature is direct: a comparison of operators that reports one run per operator, as most do, cannot distinguish a better operator from a lucky seed at the differences observed here. Five seeds were enough to show this, and to withdraw a result that three seeds had suggested (Section 3.3).

4.4 Why does the subject dominate?

The held-out subject explained 74 percent of the variance on CHB-MIT and, together with its interactions with pipeline and operator, about 60 percent on Siena. Seizures differ between people in morphology, location and duration, and the recordings in their equipment, montage quality and artefacts; a model trained on the other subjects works well or badly on a given person largely for reasons that no fusion operator or preprocessing step changes. The interaction

of subject with pipeline and operator, 7 percent on CHB-MIT and 27 percent on Siena, means that the operator that suits one person is often not the one that suits another. Averaging over subjects hides this heterogeneity, and with 23 and 14 subjects it also limits the precision of every comparison between operators: the equivalence bounds of Section 3.2 are as wide as they are because the subjects differ so much.

4.5 Does the choice of artefact removal algorithm affect reliability?

In two ways. The three algorithms changed performance differently by operator (Section 3.5), and on Siena they reordered the operators more than the seed did. Less expected, they differed in reproducibility. With the same code, package versions and seeds, GEDAI cleaned about a quarter of recordings differently on a second machine, and Infomax changed its component decision in 2.5 percent of recordings between machines and in about 1 percent between two runs on one machine; scoring the same trained models on the two cleanings changed seizure-level sensitivity by up to 0.054 and the order of the operators. Fixed seeds and pinned versions are the reproducibility measures a methods section usually reports; for ICA-based cleaning they are not sufficient, because decisions made at a threshold (a component's |z| against an EOG proxy, GEDAI's noise threshold) amplify rounding differences between machines. A perturbation of the input of one part in ten million reproduced the differences, so no machine gives the correct result and none can be chosen to avoid them. The window-level results of this study rest on one cleaning and are unaffected; but a study that cleans its training data on one machine and its test data on another, or that compares its numbers with another group's, can see differences of the size of the effect it studies. We recommend that studies using artefact removal store the cleaned signals, or at least the decomposition and the removed components, alongside the code.

4.6 Do results from one dataset carry over to another?

Performance did, rankings did not. Models trained on CHB-MIT reached on Siena the macro F1 of models trained on Siena itself, but the order of the operators on CHB-MIT, on Siena and after transfer were unrelated (mean Kendall tau between -0.11 and 0.07). Score fusion is the clearest case: first in 82 percent of Siena's own runs, it was first in 11 percent of the transfer runs and among the weaker operators on CHB-MIT. Its advantage on Siena is therefore a property of training and testing on Siena, possibly of how its blending weight is chosen on Siena's validation subjects, not a property of the operator. The two datasets differ in age, seizure types, recording protocol and mains frequency (Section 2.1), so we do not attribute the difference to any one of these.

4.7 Statistical significance, practical size and ranking stability

The three properties we kept apart gave different answers, and reporting any one alone would have misled. On Siena score fusion was first in four of five runs, a stable ranking, yet its advantage was not significant and could not be shown to exceed or fall below 0.05 macro F1. On CHB-MIT the interaction was significant before correction and with three seeds, yet its practical size was 0.004 macro F1. And on CHB-MIT several pairs of operators were practically equivalent while their order changed from seed to seed, which is what equivalence implies: when two operators are equally good, their order is decided by noise. A stable ranking is not evidence of a meaningful difference, a significant effect is not evidence of a large one, and the absence of a significant difference is not evidence of equivalence.

4.8 What this means for comparisons of fusion operators

We did not find a fusion operator that is better than the others across preprocessing, seeds and datasets, and our results do not support claims that one is. That is itself the main finding: a claim of superiority based on one dataset, one pipeline and one training run, the common design in this literature, would have reached a different conclusion in different runs of our study. Comparisons of this kind should report several seeds and, where feasible, several preprocessing choices; test differences with corrected and equivalence tests rather than naive paired tests; report the rank distribution rather than one ranking; and state the margin below which a difference is considered irrelevant before looking at the results.

4.9 What do the results mean clinically?

(Draft. P0 to P5 on CHB-MIT; the numbers for P6 follow when its event metrics are complete. Statements marked [ref] need a source from the literature review.)

None of the systems studied here is close to clinical use, and the study was not designed to produce one: the threshold was fixed at 0.5, there was no post-processing, and the models were trained on other people's seizures only. At that operating point the operators detected 68 to 80 percent of CHB-MIT's seizures with 5.3 to 7.8 false alarms per hour, that is 127 to 187 false alarms per day of monitoring, and about 2 percent of all detections were seizures. Clinically acceptable systems are usually described with false alarm rates below about one per hour, often far lower, at a high sensitivity [ref]; no person in CHB-MIT reached a mean below one false alarm per hour. Post-processing (smoothing consecutive windows, a minimum event duration) and a threshold chosen per patient typically trade a little sensitivity for far fewer false alarms [ref]; we left both out on purpose, so that the operators are compared on their raw output, and the absolute numbers should be read in that light.

Three results nevertheless bear on clinical use. First, the operators that differed by a few hundredths of macro F1 at the window level differed by little more at the seizure level, except early fusion, which detected the most seizures (0.80) at the most false alarms (7.8 per hour); which of these is preferable is a clinical choice between missed seizures and alarm fatigue [ref], not a property that window-level metrics reveal. Second, the person mattered more than any modelling choice: mean sensitivity ranged from 0.23 (chb06) to 1.00 across persons and the false alarm rate from 1.2 to 12.3 per hour, while the operators' means spanned 0.68 to 0.80 and 5.3 to 7.8. For a patient, whether the detector works depends far more on who they are than on the fusion operator, which supports patient-specific adaptation over the search for a better generic operator [ref]. Third, an artefact removal step that does not reproduce across machines (Section 4.5) is a clinical risk as much as a scientific one: the same model could detect a different set of a patient's seizures depending on where the recording is cleaned.

Siena cannot answer the clinical question of false alarms: its recordings are cut around seizures, so we report only its sensitivity (83 to 86 percent), which is consistent with CHB-MIT's and says nothing about specificity in continuous monitoring. Neither dataset contains the long, artefact-rich ambulatory or intensive-care recordings in which seizure detectors are used, and the persons are few (23 and 14).

4.10 Limitations

(draft; to be completed with the results)

- **Sample size.** Twenty-three CHB-MIT and fourteen Siena subjects make the corrected confidence intervals wide: differences between operators that are small can be neither

shown nor excluded at the equivalence margins, so many pairs are neither different nor demonstrably equivalent.

- **One design family.** All operators share one pair of representations (raw waveform and log-magnitude STFT), one backbone family and one parameter budget; other representations (wavelets, other time-frequency transforms) or larger models may behave differently.
- **Windows, not sessions.** Window-level metrics use non-ictal windows subsampled to four per ictal window, so they do not reflect the true prevalence of seizures; the seizure-level metrics on the full recordings address this but are secondary. A window is labelled ictal if it overlaps a seizure at all; excluding partly ictal windows is implemented but was not run.
- **Siena.** Its recordings are cut around seizures, so false alarms per hour cannot be interpreted and are not reported; the summed recording time (141.0 h by the EDF headers) differs from the dataset description (128.4 h) for reasons that remain unexplained for three patients. Six problems in its seizure lists were resolved by decisions documented in Supplementary Table S4.
- **CHB-MIT.** Thirteen of its 198 seizures lie in recordings without the standard montage and are not used; chb24 is not listed in the subject information file and is treated as a distinct person.
- **Artefact removal.** The settings (16 components, frontopolar channels as EOG proxies, no muscle rejection) and the projection of GEDAI's reference onto a bipolar montage are our choices for montages without EOG; GEDAI was designed for referential, high-density recordings. Infomax's and GEDAI's results do not reproduce across machines (Section 2.7); window-level results rest on one cleaning, but the seizure-level metrics of P6a differ from the training-time cleaning in 8 of 670 recordings.
- **Device.** All multiverse runs used one GPU; GPU and CPU results differ at the level of a seed change, so absolute numbers are not interchangeable with the CPU results of earlier drafts.
- **Seeds.** Five seed sets bound the seed variance only roughly; the rank analyses report it rather than average it away.
- **Literature.** Title and abstract screening of the systematic search was done by an AI assistant (agreement with an independent AI pass, kappa 0.81); a human screening and the full texts of the remaining candidates are in progress, and claims marked [abstract] in Section 1 rest on abstracts.

Use of AI assistance

An AI assistant (Claude, Anthropic) was used to help write code and to edit the text. The authors designed the study, made every methodological decision, checked the code, results and text, and take full responsibility for them.

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Supplementary Material

Table S1: Artefact removal in P6: signal retention and compute. Per recording of CHB-MIT with selected windows (670), on the 0.5 to 40 Hz signal. Power kept = signal power after removal / before. Extended Infomax and AMICA share 16 components, fitting on every 8th sample and the same automatic ocular component rule; GEDAI selects what to remove by its own criterion. Compute on a laptop CPU (Intel Core Ultra 7 155U), six recordings in parallel.

method	recordings	power kept, median	IQR	min	share < 0.5	components removed, mean	seconds per recording, median	CPU hours, total
P6a Extended Infomax	670	0.784	0.531-0.899	0.090	0.222	1.491	33	8.5
P6b GEDAI, conservative (used)	670	0.999	0.717-0.999	0.041	0.206		18	4.7
P6c AMICA	670	0.784	0.535-0.885	0.076	0.228	1.531	213	73.1
GEDAI, default (not used)	670	0.658	0.317-0.999	0.041	0.433		22	5.6

Table S2: GEDAI preset, chosen on signal preservation only. Six CHB-MIT subjects (chb01, chb03, chb05, chb08, chb12, chb15), the same windows under both presets. Power kept per recording and per window (relative to the same 0.5 to 40 Hz signal); no classification result was used for the choice (`src/gedai_qc.py`).

preset	unit	n	p5	p25	median	p75	share < 0.5	share < 0.1
default (auto)	recording	199	0.197	0.324	0.843	0.999	0.387	0.000
default (auto)	ictal window	591	0.241	0.367	0.779	1.000	0.333	0.005
default (auto)	non-ictal window	2364	0.263	0.395	0.999	0.999	0.328	0.003
conservative (auto-), used	recording	199	0.263	0.957	0.999	0.999	0.166	0.000
conservative (auto-), used	ictal window	591	0.313	0.999	1.000	1.000	0.134	0.000
conservative (auto-), used	non-ictal window	2364	0.331	0.999	0.999	1.000	0.126	0.002

Table S3: P4: technical artefact rejection per person. A window is rejected if a channel is flat (standard deviation below 1 μV) or holds one constant value for at least 0.5 s, on the unfiltered signal. Rejected windows are removed from training and validation only; test windows are unchanged. Rejection rates are given for ictal and non-ictal windows separately. chb01 includes chb21, the same person.

person	windows	ictal	ictal rejected	non-ictal	non-ictal rejected	ictal rejected %	non-ictal rejected %
chb01	370	74	0	296	1	0.0	0.3
chb02	100	20	2	80	2	10.0	2.5
chb03	225	45	7	180	1	15.6	0.6
chb04	210	42	5	168	5	11.9	3.0
chb05	310	62	0	248	0	0.0	0.0
chb06	120	24	1	96	0	4.2	0.0
chb07	180	36	2	144	1	5.6	0.7
chb08	485	97	4	388	3	4.1	0.8
chb09	155	31	3	124	0	9.7	0.0
chb10	245	49	11	196	4	22.4	2.0
chb11	415	83	1	332	0	1.2	0.0

person	windows	ictal	ictal rejected	non-ictal	non-ictal rejected	ictal rejected %	non-ictal rejected %
chb12	615	123	1	492	7	0.8	1.4
chb13	335	67	0	268	5	0.0	1.9
chb14	120	24	0	96	0	0.0	0.0
chb15	1070	214	0	856	1	0.0	0.1
chb16	70	14	0	56	1	0.0	1.8
chb17	160	32	0	128	6	0.0	4.7
chb18	190	38	1	152	0	2.6	0.0
chb19	135	27	3	108	0	11.1	0.0
chb20	180	36	1	144	2	2.8	1.4
chb22	120	24	0	96	0	0.0	0.0
chb23	245	49	0	196	0	0.0	0.0
chb24	325	65	0	260	0	0.0	0.0
all	6380	1276	42	5104	39	3.3	0.8

Table S4: Siena: metadata problems and the decisions taken. Every departure from the published seizure lists, with the decision used in the main analysis, its kind (a correction verified against the recording, or a decision rule applied without such a check), how it was checked, and how much the alternative would change the windows. Other, purely textual fixes (a space inside a time in PN10's list, misspelt file names, PN12's missing registration times taken from the same file's first block, recordings past midnight read modulo 24 h) are listed in `docs/SIENA.md`, Section 4. Recording time is taken from the EDF headers, 141.0 h; the dataset's description gives 128.4 h, a difference explained for PN14 (the start time above) but not for PN03, PN10 and PN12, whose files contain no padding (`docs/SIENA.md`, Section 8).

patient	recording / seizure	problem in the metadata	used	type	how it was checked	remaining uncertainty	effect on the windows
PN00	PN00-3, seizure 3	listed end 19.29.29 lies 1916 s after the recording ends (18.57.33)	end read as 18.29.29 (a 60 s seizure)	verified correction	signal: the right fron-totemporal 3-20 Hz ictal rhythm (FP2-F8, F8-T8) falls below a quarter of its peak 14 s after 18.29.29; the patient's other four seizures end 0 to 13 s after their listed ends by the same rule	offset from an automatic rule on one channel pair	alternative (exclude the seizure and the rest of the recording): 30 instead of 37 ictal windows for PN00
PN05	PN05-3, seizure 3	registration start: list 06.01.23, EDF header 06.01.13	EDF header start time, as for every recording	decision rule (not a verified correction)	none possible: the header defines the first sample and 39 of 41 recordings agree with their list; the list's value fits the 20 s end offset of 18 other recordings, the header's does not	10 s in the seizure's position	4 ictal windows either way, shifted by one: 2 window labels differ

patient	recording / seizure	problem in the metadata	used	type	how it was checked	remaining uncertainty	effect on the windows
PN10	PN10-3, seizure 3	two onsets: 15.43.53 (clinical), 15.43.59 (electrical)	electrical onset (advisor)	decision	CHB-MIT annotates electrographic onsets; the electrical onset keeps the two corpora consistent	none (both listed)	clinical onset instead: 48 instead of 47 ictal windows for PN10 (1 label)
PN10	PN10-4.5.6, seizure 6	only a clinical onset (15.18.26)	used as listed, flagged in the corpus's seizure notes	decision	-	electrical onset unknown; in seizure 3 it followed the clinical one by 6 s	at most about one window, by analogy with seizure 3
PN10	PN10-2, seizure 2	end "11.41.04 opure 11.40.43" (Italian "or")	first value, 11.41.04	decision rule	-	21 s in the seizure's end	second value instead: 45 instead of 47 ictal windows for PN10
PN14	PN14-3	registration start: list 16.17.45, EDF header 19.17.45	EDF header start time	verified correction	header start + duration (41,995 s) gives the listed end time 06.57.40 exactly; the list's start would make the recording three hours longer than the file	none	seizure at 1 h 52 min into the recording, not 4 h 52 min

Table S5: Artefact removal: power kept per window, by band and class. What each method removes from the 0.5 to 40 Hz signal, per 10 s window (cleaned power divided by the power of the same window before cleaning, summed over channels), for ictal and non-ictal windows, overall and in five bands (delta 0.5-4, theta 4-8, alpha 8-13, beta 13-30, low gamma 30-40 Hz; medians). The second part compares the classes within recordings: per recording the median of each class, their difference, and a Wilcoxon signed-rank test over the recordings that hold both. A difference between the classes over the whole corpus that is absent within recordings reflects which recordings a method cleans heavily, not a preference for one class (`src/p6_signal_effects.py`). GEDAI's default preset was applied to CHB-MIT only.

dataset	method	windows	n	power kept, median	share < 0.5	delta	theta	alpha	beta	low gamma
chbmit	Extended Infomax	ictal	1276	0.872	0.092	0.855	0.905	0.929	0.933	0.935
chbmit	Extended Infomax	non-ictal	5104	0.854	0.187	0.835	0.893	0.911	0.927	0.933
chbmit	GEDAI conservative (used)	ictal	1276	1.000	0.134	0.999	1.000	1.000	1.000	1.000

dataset	method	windows	n	power kept, median	share < 0.5	delta	theta	alpha	beta	low gamma
chbmit	GEDAI conservative (used)	non-ictal	5104	0.999	0.167	0.999	1.000	1.000	1.000	1.000
chbmit	GEDAI default (not used)	ictal	1276	0.625	0.448	0.583	0.739	0.836	0.882	0.909
chbmit	GEDAI default (not used)	non-ictal	5104	0.925	0.372	0.920	0.966	0.983	0.991	0.994
chbmit	AMICA	ictal	1276	0.873	0.093	0.851	0.909	0.922	0.929	0.933
chbmit	AMICA	non-ictal	5104	0.839	0.191	0.821	0.882	0.901	0.921	0.930
siena	Extended Infomax	ictal	328	0.682	0.271	0.609	0.759	0.847	0.900	0.941
siena	Extended Infomax	non-ictal	1312	0.657	0.370	0.520	0.714	0.862	0.906	0.914
siena	GEDAI conservative (used)	ictal	328	0.679	0.427	0.694	0.745	0.850	0.833	0.944
siena	GEDAI conservative (used)	non-ictal	1312	0.997	0.383	0.997	1.000	1.000	1.000	1.000
siena	AMICA	ictal	328	0.723	0.232	0.660	0.823	0.887	0.929	0.959
siena	AMICA	non-ictal	1312	0.719	0.334	0.627	0.800	0.915	0.936	0.939

Within recordings:

dataset	method	recordings	ictal - non-ictal, median	share with ictal lower	Wilcoxon p
chbmit	Extended Infomax	137	0.0098	0.2847	0.0001
chbmit	GEDAI conservative (used)	137	0.0001	0.3212	0.0016
chbmit	GEDAI default (not used)	137	-0.0000	0.5036	0.0363
chbmit	AMICA	137	0.0170	0.2847	0.0000
siena	Extended Infomax	41	-0.0050	0.5122	0.6769
siena	GEDAI conservative (used)	41	-0.0000	0.5610	0.6815
siena	AMICA	41	0.0000	0.4390	0.8790

Table S6: Window-level performance per pipeline and operator. Mean and SD over the five seed sets of each run's fold mean, for every pipeline and fusion operator (`python -m src.make_results`, `results_v2/results/main_*.csv`). **CHB-MIT, macro F1** (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.678 ±0.013	0.679 ±0.012	0.671 ±0.013	**0.683** ±0.020	0.661 ±0.014
P1	0.684 ±0.008	0.665 ±0.015	**0.691** ±0.005	0.687 ±0.008	0.672 ±0.017
P2	0.680 ±0.014	**0.688** ±0.009	0.680 ±0.021	0.687 ±0.009	0.675 ±0.008
P3	0.679 ±0.011	0.677 ±0.017	0.676 ±0.015	**0.689** ±0.015	0.678 ±0.014
P4	0.688 ±0.010	0.678 ±0.008	**0.694** ±0.012	0.670 ±0.010	0.660 ±0.024
P5	0.678 ±0.012	0.684 ±0.019	0.690 ±0.021	**0.693** ±0.005	0.681 ±0.008
P6a	0.672 ±0.014	0.679 ±0.014	**0.692** ±0.018	0.677 ±0.013	0.675 ±0.012
P6b	0.662 ±0.012	0.686 ±0.018	**0.691** ±0.012	0.681 ±0.020	0.682 ±0.028
P6c	0.671 ±0.007	0.688 ±0.019	**0.688** ±0.013	0.679 ±0.014	0.674 ±0.010

CHB-MIT, AUPRC (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.632 ±0.016	0.656 ±0.018	**0.662** ±0.019	0.655 ±0.015	0.654 ±0.011
P1	0.623 ±0.015	0.665 ±0.011	**0.673** ±0.009	0.665 ±0.017	0.667 ±0.019
P2	0.618 ±0.021	0.668 ±0.007	**0.683** ±0.016	0.669 ±0.013	0.657 ±0.021
P3	0.607 ±0.013	0.657 ±0.012	0.655 ±0.016	**0.658** ±0.031	0.651 ±0.011
P4	0.628 ±0.018	0.672 ±0.018	**0.673** ±0.013	0.650 ±0.026	0.651 ±0.018
P5	0.608 ±0.019	0.642 ±0.017	**0.664** ±0.022	0.660 ±0.012	0.662 ±0.005
P6a	0.619 ±0.021	0.653 ±0.028	**0.668** ±0.015	0.652 ±0.017	0.657 ±0.010
P6b	0.564 ±0.010	0.635 ±0.011	**0.643** ±0.018	0.636 ±0.015	0.631 ±0.015
P6c	0.617 ±0.009	0.658 ±0.023	**0.671** ±0.010	0.654 ±0.018	0.668 ±0.012

CHB-MIT, balanced accuracy (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	**0.684** ±0.009	0.681 ±0.010	0.674 ±0.010	0.684 ±0.016	0.667 ±0.014
P1	0.684 ±0.006	0.669 ±0.010	**0.685** ±0.008	0.684 ±0.008	0.680 ±0.015
P2	0.684 ±0.010	0.682 ±0.006	0.679 ±0.022	**0.685** ±0.009	0.678 ±0.010
P3	**0.687** ±0.010	0.676 ±0.015	0.676 ±0.012	0.687 ±0.013	0.683 ±0.011
P4	**0.690** ±0.008	0.677 ±0.006	0.687 ±0.008	0.667 ±0.011	0.663 ±0.020
P5	0.686 ±0.015	0.683 ±0.014	0.684 ±0.019	**0.694** ±0.008	0.687 ±0.008
P6a	0.676 ±0.014	0.675 ±0.014	**0.688** ±0.017	0.680 ±0.006	0.679 ±0.014
P6b	0.670 ±0.009	0.682 ±0.017	**0.688** ±0.012	0.680 ±0.016	0.686 ±0.017
P6c	0.677 ±0.008	0.682 ±0.017	**0.686** ±0.009	0.680 ±0.015	0.681 ±0.010

Siena, macro F1 (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.601 ±0.021	0.616 ±0.023	0.608 ±0.016	0.598 ±0.020	**0.639** ±0.008
P1	0.616 ±0.024	0.637 ±0.012	0.623 ±0.020	0.628 ±0.026	**0.665** ±0.019
P2	0.610 ±0.016	0.639 ±0.014	0.636 ±0.018	0.625 ±0.016	**0.674** ±0.012
P3	0.613 ±0.014	0.652 ±0.039	0.620 ±0.013	0.613 ±0.010	**0.664** ±0.029
P4	0.614 ±0.020	0.637 ±0.012	0.621 ±0.019	0.627 ±0.025	**0.666** ±0.020
P5	0.637 ±0.029	0.642 ±0.017	0.650 ±0.015	0.652 ±0.008	**0.690** ±0.031
P6a	0.619 ±0.018	0.622 ±0.019	0.621 ±0.048	0.611 ±0.013	**0.672** ±0.018
P6b	0.614 ±0.010	0.681 ±0.012	**0.684** ±0.015	0.652 ±0.018	0.676 ±0.027
P6c	0.629 ±0.027	0.638 ±0.010	0.613 ±0.024	0.621 ±0.013	**0.683** ±0.026

Siena, AUPRC (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.453 ±0.018	0.556 ±0.023	0.527 ±0.018	0.522 ±0.003	**0.571** ±0.014
P1	0.498 ±0.025	0.613 ±0.039	0.615 ±0.019	0.601 ±0.021	**0.649** ±0.024
P2	0.494 ±0.036	0.624 ±0.035	0.594 ±0.019	0.599 ±0.020	**0.642** ±0.007
P3	0.492 ±0.020	0.633 ±0.043	0.611 ±0.013	0.588 ±0.032	**0.650** ±0.029
P4	0.497 ±0.026	0.613 ±0.039	0.614 ±0.017	0.601 ±0.022	**0.648** ±0.025
P5	0.540 ±0.021	0.611 ±0.027	0.610 ±0.019	0.582 ±0.023	**0.641** ±0.027
P6a	0.510 ±0.030	0.587 ±0.042	0.574 ±0.045	0.575 ±0.041	**0.621** ±0.028
P6b	0.460 ±0.020	0.590 ±0.017	0.585 ±0.022	0.571 ±0.028	**0.637** ±0.026
P6c	0.522 ±0.034	0.563 ±0.028	0.585 ±0.015	0.578 ±0.034	**0.618** ±0.034

Siena, balanced accuracy (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.621 ±0.021	0.663 ±0.018	0.652 ±0.011	0.639 ±0.013	**0.665** ±0.007
P1	0.643 ±0.022	0.676 ±0.014	0.673 ±0.018	0.673 ±0.024	**0.689** ±0.022
P2	0.645 ±0.024	0.680 ±0.008	0.684 ±0.012	0.674 ±0.014	**0.696** ±0.010
P3	0.641 ±0.018	**0.693** ±0.031	0.671 ±0.010	0.660 ±0.009	0.684 ±0.021
P4	0.643 ±0.022	0.676 ±0.014	0.673 ±0.018	0.673 ±0.025	**0.691** ±0.024
P5	0.661 ±0.023	0.688 ±0.015	0.691 ±0.011	0.687 ±0.010	**0.710** ±0.038
P6a	0.645 ±0.016	0.668 ±0.016	0.664 ±0.038	0.658 ±0.015	**0.682** ±0.011
P6b	0.622 ±0.011	0.688 ±0.017	**0.694** ±0.018	0.676 ±0.015	0.691 ±0.022
P6c	0.654 ±0.019	0.676 ±0.006	0.660 ±0.019	0.663 ±0.020	**0.697** ±0.018